The documentation and process conversion measures necessary to comply with this revision shall be completed by 5 June 93.

INCH-POUND

MIL-S-19500/428B
5 March 1993
SUPERSEDING
MIL-S-19500/428A
7 December 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTOR, N-CHANNEL SILICON,
TYPE 2N4416A
JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for N-Channel, junction, silicon field-effect transistors. Three level of product assurance is provided for the device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Types	P_T 1/ T_A = 25°C	V_{DG} and V_{DS}	V_{GS}	I_G	T_{STG} and T_J
	mW	V dc	V dc	mA dc	°C
2N4416A	300	35	-35	10	-65 to +200

1/ Derate linearly, 1.7 mW/°C for T_A = 25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	I_{DSS} 1/ V_{DS} = 15 V dc, V_{GS} = 0	$V_{GS(off)}$ V_{DS} = 15 V dc, I_D = -1.0 nA dc	F	
			V_{DS} = 15 V dc, I_D = 5 mA dc, R_G = 1 k ohms, f = 100 MHz	V_{DS} = 15 V dc, I_D = 5 mA dc, R_G = 1 k ohms, f = 400 MHz
	mA dc	V dc	dB	dB
Minimum	5.0	-2.5		
Maximum	15.0	-6.0	2	4

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_A = 25°C - Continued

Limit	$\lvert y_{fs} \rvert$ 2/ V_{DS} = 15 V dc V_{GS} = 0 f = 1 kHz	$\lvert y_{os} \rvert$ 2/ V_{DS} = 15 V dc V_{GS} = 0 f = 1 kHz	C_{oss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz	C_{iss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz	C_{rss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz
	ms	µS	pF	pF	pF
Minimum	4.5				----
Maximum	7.5	50	2	4	0.8

2/ Pulsed width = 100 ms; duty cycle = 10%.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

R_G' - - - - - - - - - - - - - - Transformed equivalent source resistance.

3.3 Design, construction, and physical dimensions. The design, construction and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

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3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9 and 10	Not applicable	Not applicable
11	I_{GSS1}, I_{DSS}, and $\|y_{fs}\|_2$	I_{GSS1}, I_{DSS}, and $\|y_{fs}\|_2$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{DSS} = 10 percent of initial value. $\Delta\|y_{fs}\|_2$ = 20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{DSS} = 10 percent of initial value. $\Delta\|y_{fs}\|_2$ = 20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = +175°C; V_{GS} = -24 V dc; V_{DS} = 0 V dc.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 3: Condition for bond strength is test condition A.

b. Subgroup 4. Condition for intermittent operation life are as follows: P_T = 300 mW at T_A = +25°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling shall be permitted.

c. Subgroup 5: Condition for bond strength (AL-Au die interconnects only) is test condition A.

d. Subgroup 5: Condition for accelerated steady-state life are as follows: V_{DS} = 15 V dc; I_D = 20 mA dc at T_A = 100°C or adjusted as required by the chosen T_A to achieve an average lot T_J = +275°C.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 3: Condition for steady-state operation life is V_{GS} = -24 V dc; V_{DS} = 0; T_A = +175°C ±3°C. No heat sink or forced-air cooling on the devices shall be permitted.

b. Subgroup 4: Condition for bond strength is test condition A.

c. Subgroup 6: Condition for high-temperature life (nonoperating) is T_A = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500. Subgroup 6: Condition for steady-state operation life is V_{GS} = -24 V dc; V_{DS} = 0; T_A = +175°C ±3°C. No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Disposition of case lead during electrical measurements. All electrical measurements and operating life test shall be performed with the case lead connected to the source.

4.5.3 Small-signal common-source short-circuit input, output, or forward transfer conductance and susceptance. These tests shall be conducted with a General Radio transfer function and admittance bridge model 1607A (or suitable equivalent) in accordance with the portion of its accompanying handbook which is applicable to this measurement. A Hewlett-Packard generator model 608D and Nems-Clarke receiver model 1502A (or suitable equivalents) shall be used with the transfer function and admittance bridge.

4.5.4 Spot noise figure tests. These tests shall be conducted using the equipment and circuit shown on figures 3 and 4, or suitable equivalent.

4.5.5 Small-signal common-source insertion power gain. This test shall be conducted using the circuit shown on figure 4, or suitable equivalent.

4.5.6 Small-signal common-source short-circuit output capacitance. This test shall be conducted using the circuit shown on figure 5, or suitable equivalent.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, gate to source	3401	Bias condition C; I_G = 1 µA dc; V_{DS} = 0	$V_{(BR)GSS}$	-35		V dc
Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc; V_{DS} = 0	I_{GSS1}		-0.1	nA dc
Drain current	3413	Bias condition C, V_{DS} = 15 V dc; pulsed (see 4.5.1)	I_{DSS}	5	15	mA dc
Gate to source voltage	3403	V_{DS} = 15 V dc; I_D = 0.5 mA dc	V_{GS}	-1	-5.5	V dc
Gate to source cutoff voltage	3403	V_{DS} = 15 V dc; I_D = 1.0 nA dc	$V_{GS(off)}$	-2.5	-6.0	V dc
Gate to source forward voltage	3403	V_{DS} = 0 V dc; I_G = 1.0 mA dc	V_{GSF}		1	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc;	I_{GSS2}		-0.1	µA dc
Low-temperature operation		T_A = -55°C				
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs}\|1$		11.25	mS
Subgroup 4						
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz; pulse width = 100 ms; duty cycle = 10 percent	$\|y_{fs}\|2$	4.5	7.5	mS

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal, common-source short-circuit input capacitance	3431	V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz	C_{iss}		4	pF
Small-signal, common-source short-circuit reverse transfer capacitance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz	C_{rss}		0.8	pF
Small-signal, common-source short-circuit output capacitance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz (see 4.5.6)	C_{oss}		2	pF
Small-signal, common-source short-circuit input susceptance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; (see 4.5.3)	b_{is}			
Test 1		f = 100 MHz			2.5	mS
Test 2		f = 400 MHz			10	mS
Small-signal, common-source short-circuit output susceptance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; (see 4.5.3)	b_{os}			
Test 1		f = 100 MHz			1	mS
Test 2		f = 400 MHz			4	mS
Small-signal, common-source short-circuit input conductance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; (see 4.5.3)	g_{is}			
Test 1		f = 100 MHz			0.3	mS
Test 2		f = 400 MHz			3.0	mS
Small-signal, common-source short-circuit output conductance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; (see 4.5.3)	g_{os}			
Test 1		f = 100 MHz			75	µS
Test 2		f = 400 MHz			400	µS

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal, common-source short-forward transfer conductance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 400 MHz (see 4.5.3)	g_{fs}	4		mS
Small-signal, common-source insertion power gain		V_{DS} = 15 V dc; I_D = 5 mA dc; R_G = 1 kΩ (see 4.5.5)	G_{ps}			
Test 1		f = 100 MHz		18		dB
Test 2		f = 400 MHz		10		dB
Common-source spot capacitance		V_{DS} = 15 V dc; I_D = 5 mA dc; R_G = 1 kΩ (see 4.5.4)	F			
Test 1		f = 100 MHz			2	dB
Test 2		f = 400 MHz			4	dB
Subgroup 5						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc; V_{DS} = 0	I_{GSS1}		0.1	nA dc
2	Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc; V_{DS} = 0	I_{GSS1}		1.0	nA dc
3	Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz; t_p = 100 ms; duty cycle = 10%	$\|y_{fs}\|_2$	4.5	7.5	mS
4	Drain current	3413	Bias condition C; V_{DS} = 15 V dc pulsed (see 4.5.1)	I_{DSS}	5	15	mA dc
5	Breakdown voltage gate to source	3401	Bias condition C; I_G = 1.0 µA dc V_{DS} = 0	$V_{(BR)GSS}$	-35		V dc
6	Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz; t_p = 100 mS; duty cycle = 10%	$\Delta\|y_{fs}\|_2$	20 percent change from initial value		
7	Drain current	3413	Bias condition C; V_{DS} = 15 V dc pulsed (see 4.5.1)	ΔI_{DSS}	10 percent change from initial value		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, and 5.

b. Subgroup 4, see table II herein, steps 1, 3, 4, 5, and 6.

c. Subgroup 5, see table II herein, steps 1, 3, 4, 5, 6, and 7.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

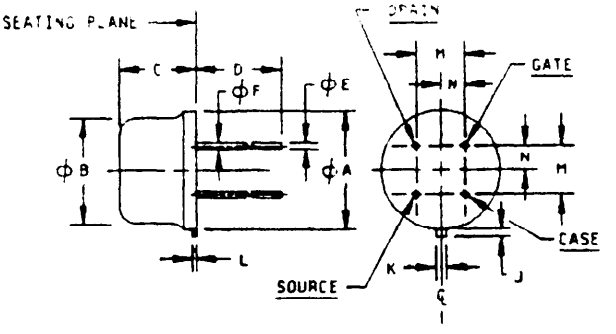
b. Subgroup 3 and 6, see table II herein, steps 2, 3, and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2 and 3, see table II herein, steps 1, 3, 4, and 5 for JANS level and steps 1 and 3 for JANTX and JANTXV levels.

b. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, and 7 for JANS level and steps 2, 3, and 6 for JANTX and JANTXV levels.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕA	.209	.230	5.31	5.84	
ϕB	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	.750	12.70	19.05	7
ϕE		.021		0.53	2,7
ϕF	.016	.019	0.41	0.48	3,7
J	.028	.048	0.71	1.22	6
K	.036	.046	0.91	1.17	
L		.020		0.51	
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, - .000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred measurement method.
5. The gate shall be electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All four leads.

FIGURE 1. Physical dimensions.

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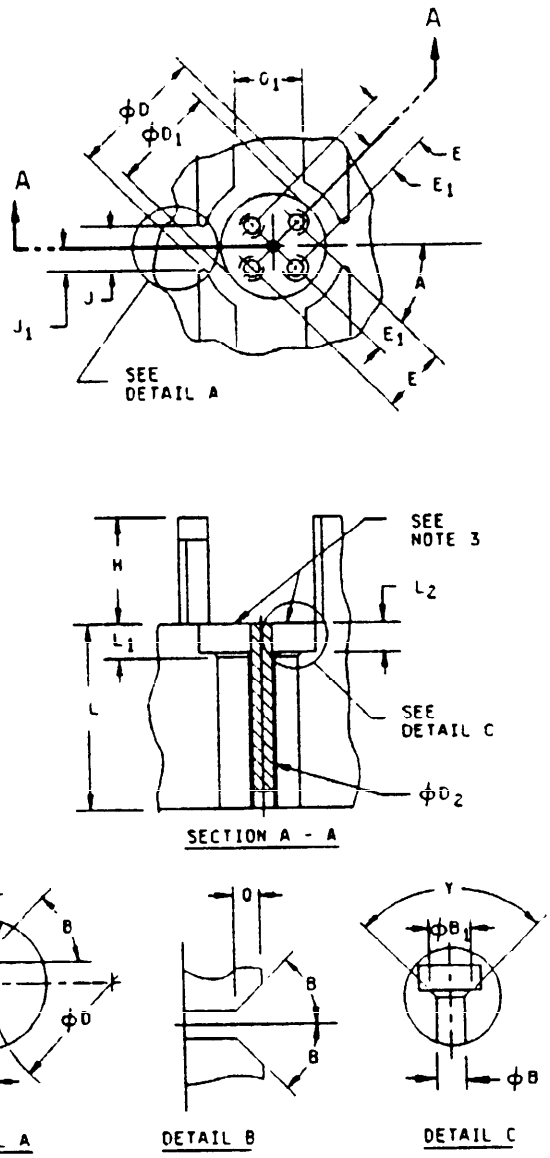


FIGURE 2. Gauge for lead and tab location for transistor type 2N4416A.

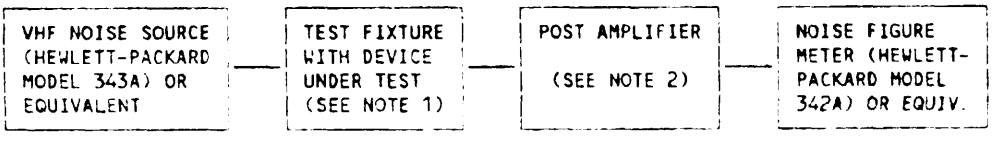
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0325	.0335	0.83	0.85	4
ϕB_1	.043 Nom		1.09 Nom		4
ϕD	.2310	.2315	5.87	5.88	
ϕD_1	.159	.161	4.04	4.09	
ϕD_2	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.53	2.55	
E_1	.0495	.0505	1.26	1.28	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.19	1.21	
J_1	.0235	.0245	0.60	0.62	
K	.009	.011	0.23	0.28	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°			
B	45° Nom				
Y	90° Nom				

NOTES:
1. Metric equivalents are given for general information only.
2. The following gauge procedure shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) +.010 (0.25 mm) from seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not to be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. a spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gauge for lead and tab location for transistor type 2N4416A - Continued.

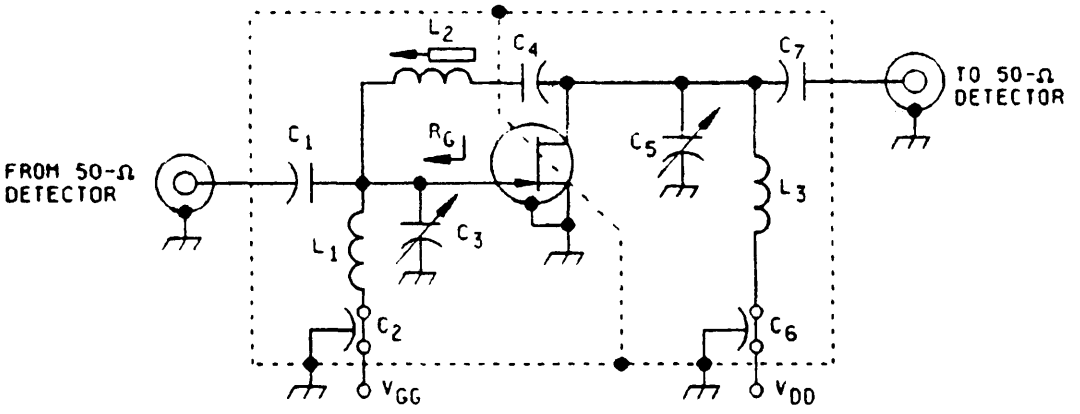
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NOTES:
1. The test fixture and procedure shall be in accordance with figure 4.
2. The post amplifier shall contain suitable biasing circuits and shall be constructed utilizing very high frequency design technique. It shall be designed for the test frequency.

FIGURE 3. Block diagram for spot noise figure tests.

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Circuit component information (See note)					
	Capacitors			Inductors	
	100 MHz	400 MHz		100 MHz	400 MHz
C_1 C_2	7 pF 0.0015 µF	1.8 pF 0.001 µF	L_1	0.14 µH, 3.5 T, no. 18 enameled copper wire, 3/8" I.D., 1/4" long	0.022 µH, 5/8" of no. 16 copper wire formed to 0.5 T, 1/4" I.D.
C_3 C_4 C_5	1-12 pF 1000 pF 1-12 pF	0.8-8 pF 27 pF 0.8-8 pF	L_2	3 µH, 17 T, no. 28 enameled copper wire, close wound, 9/32" I.D., powered iron slug	0.2 µH, 6 T, no. 24 enameled copper wire, close wound, 7/32" I.D., aluminum slug
C_6 C_7	0.0015 µF 3 pF	0.001 µF 1 pF	L_3	0.25 µH, 4.5 T, no. 18 enameled copper wire, 3/8" I.D., 5/16" long	0.03 µH, 1 1/2" of no. 16 enameled copper wire formed to 1 T, 3/8" I.D.

NOTE: Transformed equivalent source resistance is 1,000Ω at 100 MHz for 100 MHz amplifier, and 1,000Ω at 400 MHz for 400 MHz amplifier.

FIGURE 4. Neutralized small-signal common-source insertion power gain and common-source spot noise figure test circuit.

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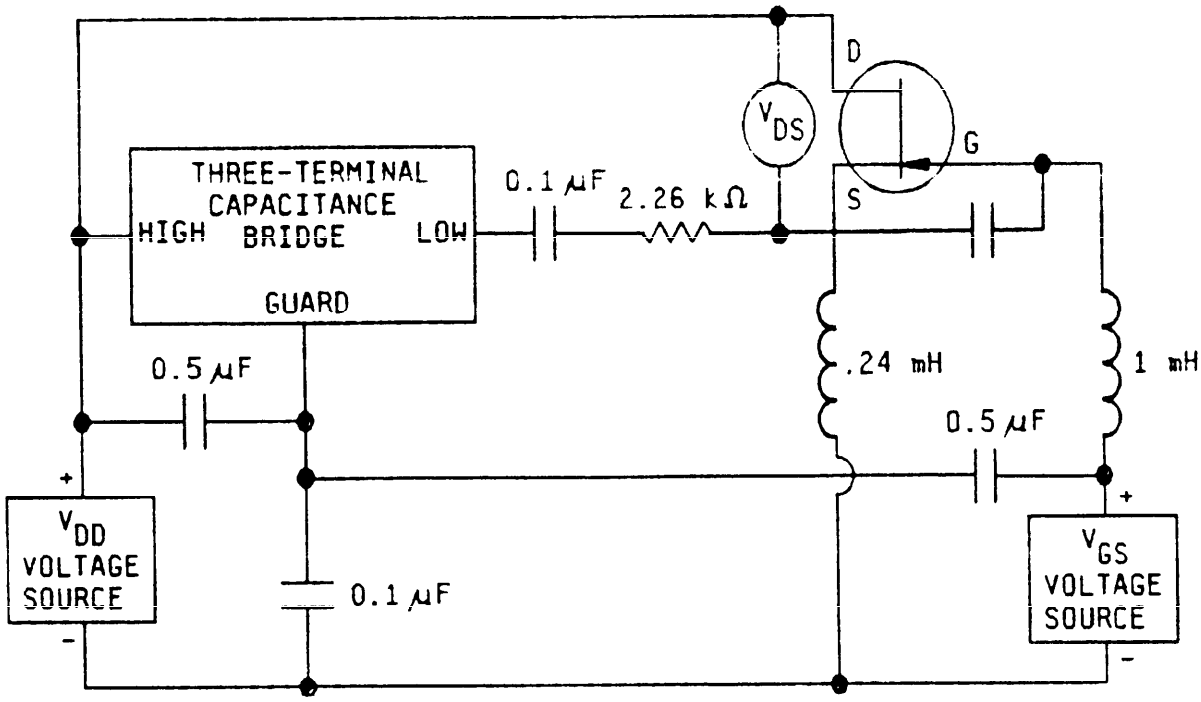


FIGURE 5. Small-signal common-source short-circuit output capacitance test circuit

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17
Navy - EC
NASA - NA

Review activities:
Navy - MC
Air Force - 85, 99
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1418)

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